



ONLINE EXAMINATION & PROCTORING PLATFORM

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Abstract

- Online Examination & Proctoring Platform is a web-based system designed to conduct and manage online examinations securely. It provides separate interfaces for students and faculty, exam creation and management, online test submission, automatic evaluation, results, and real-time proctoring. The system uses React, Node.js, Express.js and MongoDB.

Problem Statement & Objectives

- Traditional and basic online examination systems have limited monitoring, manual examination management and insufficient mechanisms for detecting suspicious activities during online tests.
- Develop a centralized platform for creating, conducting and managing online examinations.
- Implement automated proctoring features to monitor student activities during examinations.

Existing vs Proposed System

Feature	Existing System	Proposed System
Exam Management	Manual / limited	Centralized digital management
Evaluation	Manual or basic	Automatic evaluation
Proctoring	Limited Monitoring	Camera-based activity monitoring
Results	Manual processing	Automatic result generation
[Limitation / Gap]	Limited monitoring during online exams	Integrated proctoring and activity logging

Feasibility Study

- Technical:** Technically feasible using React, Node.js, Express.js, MongoDB and REST APIs with commonly available development tools.
- Economic:** Low-cost implementation using open-source technologies and locally available development tools.
- Operational:** Simple web-based interface with separate workflows for students and faculty.
- Schedule:** Designed and implemented within the semester project timeline.

Results & Testing

The system successfully supports user registration and authentication, role-based access, examination creation, question management, online examination, automatic evaluation, result generation and proctoring activity monitoring.

Conclusion

The Online Examination & Proctoring Platform provides an integrated solution for conducting and managing online examinations. It combines examination management, automated evaluation, result generation and proctoring features in a single web-based platform.

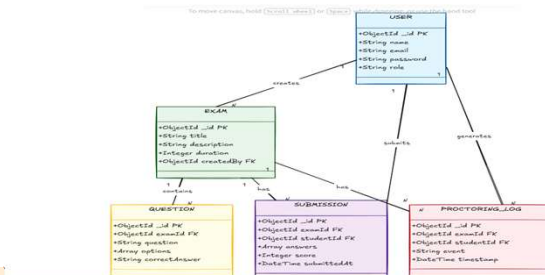
Tools & Technologies

- Front-End: React.js, HTML, CSS, JavaScript
- Back-End: Node.js, Express.js, REST APIs
- Database: MongoDB
- Tools: VS Code, Git & GitHub

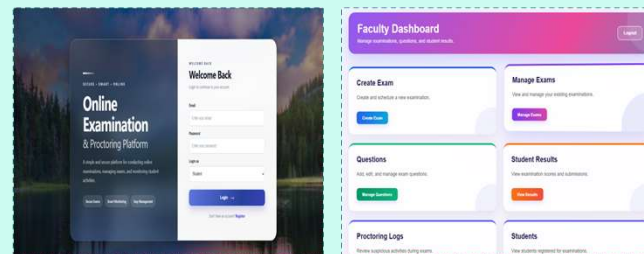
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

Improved analytics and performance reports

Cloud deployment and scalable infrastructure

